Tezpur University
Spring Semester Mid examination-2025.

Course Code: CSMT534 | **Course Title: Data Science**

Full marks=30 | *Time = 1.5 hours*

(The figures in the right-hand margin indicate full marks for the questions)

****All questions are compulsory**

Q1. Explain the difference between supervised and unsupervised learning with suitable examples. Also discuss when each approach is preferred in data science applications. 3+2=5

Q2. Describe the concept of feature selection. Why is it important in machine learning models and what could happen if it is ignored? 2+3=5

Q3. Consider the records of Student X and Student Y from the dataset given below. Compute the similarity between Student X and Student Y using a suitable method for mixed data (e.g.. Gower-based approach) 5

Attribute	Type	Student X	Student Y
GPA	Numeric	7.8	8.5
Study Hours/Week	Numeric	12	15
Mode of Study	Categorical	Online	Offline
Scholarship Holder	Binary	No	Yes

Q4. Consider the binary classification dataset with 3 features X1, X2 and X3 and 6 samples. Select the two best features according to the F-score computation. 10

Sample	Class	X1	X2	X3
1	1	9	4	2
2	1	8	5	3
3	1	10	3	1
4	-1	2	6	8
5	-1	3	7	7
6	-1	1	5	9

Answer either Question Q5 (a) OR Q5 (b).

Q5.(a) Critically evaluate whether the similarity score of question Q3 above provides a meaningful basis for grouping the students in a cluster. Justify your conclusion by commenting on the influence of each attribute on the final similarity. 5

OR

Q5.(b) You are given the following 10 two-dimensional data points: A(1,2), B(2,1), C(1,1), D(8,8), E(9,8), F(8,9), G(5,5), H(5,6), I(6,5), J(50,50). DBSCAN is executed twice with the following parameter settings. Evaluate which parameter setting (Run 1 or Run 2) is more appropriate for this dataset. 5

Run 1: ε=2, minPts = 3; **Run 2:** ε=5, minPts = 2

1. Compare any two search strategies for generating the best feature subset optimally. [2]

2. You are a data scientist working with a dataset that includes a feature named `Customer_Credit_Value` (CCV). The raw data for this attribute from a small sample of 10 customers is as follows: 150,200,220,250,280,300,350,400,450,2000 [4]
   a. Calculate the standardized value (Z-score) for the outlier value of 2000.
   b. Calculate the normalized value (Min-Max scaling) for the same outlier value of 2000.

Based on your calculations, **evaluate** and **justify** which of these two scaling methods (Standardization or Normalization) is more suitable for this specific dataset. Your answer should explicitly discuss how the **outlier** of 2000 impacts each scaling method.

3. You are a data analyst for a mobile gaming company running an A/B test to determine if a new app layout increases the conversion rate for in-app purchases. The null hypothesis is that the new layout has no effect on the conversion rate. After one week, you collect the following results: [4]

	Made a Purchase	**Did not Make a Purchase**	**Total**
Group A(Current Layout):	100	2,400	**2,500**
Group B (New Layout):	125	2,375	**2,500**
Total	**225**	**4,775**	**5,000**

Perform a chi-square test for independence with a significance level (α) of 0.05, the critical value for this test is approximately 3.84. Should the company implement the new layout? Justify your recommendation.

ORG-17

Tezpur University
Spring Semester End examination-2025.

Course Code: CSMT534
Full marks=30

Course Title: Data Science
Time = 2 hours

(The figures in the right-hand margin indicate full marks for the questions)

Q1. A binary classification dataset contains 12 samples with the following class distribution: 1+4+5=10

Class 0: 5 samples

Class 1: 7 samples

A decision tree is evaluating a split that produces two child nodes:

Left Node: 4 samples (3 of Class 0, 1 of Class 1)

Right Node: 8 samples (2 of Class 0, 6 of Class 1)

(a) Compute the entropy of the parent node.

(b) Compute the entropy of each child node and the information gain for the split.

(c) Explain how bagging would train multiple classifiers using this dataset, and describe how AdaBoost would update sample weights after a weak classifier misclassifies 4 out of 12 instances.

Q2. A dataset has 10 points labeled P1 to P10. The pairwise distances between points are shown below. OPTICS is run with $\varepsilon = 2.0$ and MinPts = 3. 5+5=10

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
P1	–	1.2	1.0	4.8	5.0	5.1	8.0	8.3	8.2	8.5
P2		–	1.1	5.0	4.9	5.3	8.2	8.0	8.4	8.6
P3			–	4.9	5.2	5.4	8.1	8.2	8.5	8.7
P4				–	1.3	1.2	6.1	6.0	6.4	6.2
P5					–	1.4	6.0	6.3	6.1	6.5
P6						–	6.2	6.1	6.6	6.4
P7							–	0.9	1.0	1.2
P8								–	0.8	1.3
P9									–	1.1
P10										–

(a) For point P4, compute the core distance using MinPts = 3. Based on the structure of the distances, determine how many clusters OPTICS is likely to detect and justify using reachability-distance reasoning.

(b) Justify how the reachability plot for this dataset would visually indicate cluster boundaries.

Q3. Consider the following transaction database with a minimum support count = 3. 8+2=10

T1: {B, A, D, C} T2: {B, C, E} T3: {A, B, C, E}

T4: {B, D, E} T5: {A, C, D} T6: {B, C, D, E}

(a) Construct the FP-tree and analyse why sorting itemsets in descending frequency order helps FPtree construction. For the item D, write down its conditional pattern base and list the items that would appear in its conditional FP-tree.

(b) Analyze a scenario where D's conditional FP-tree would not contain any branches (i.e., collapses to empty).